

Pumping Station IoT — Documentation

Project: Remote Pump Monitoring & Control System **Hardware:** Waveshare ESP32-S3 A7670E 4G · PZEM-004T AC Power Meter **Cloud:** Google Firebase (Realtime DB + Firestore + Cloud Functions)

Document Index

#	Document	Audience	Description
1	System Overview	Everyone	What the system does, components, architecture
2	Hardware Setup & Wiring	Technician / Engineer	PZEM-004T wiring, board, SIM, antenna
3	Firmware & Provisioning	Developer / Technician	Flash firmware, register new stations
4	Dashboard User Guide	Operator / Client	Login, map, charts, alerts, history export
5	Backend & Cloud Setup	Developer	Firebase project, deploy Cloud Functions
6	Alerts & Scaling Guide	Developer / Operator	Threshold config, adding 400+ stations
7	Local Development Guide	Developer	Emulators, seeding, simulator setup

Quick Start

1. **New to the system?** → Start with [System Overview](#)
2. **Setting up hardware?** → Go to [Hardware Setup & Wiring](#)
3. **Adding a new station?** → Follow [Firmware & Provisioning](#)
4. **Using the web dashboard?** → See [Dashboard User Guide](#)
5. **Running locally?** → Follow [Local Development Guide](#)

For support, contact the development team.